

1-form.

Def. A map $\omega: \bigcup_{p \in \mathbb{R}^3} T_p \mathbb{R}^3 \rightarrow \mathbb{R}$ is called a 1-form if: for fixed $p \in \mathbb{R}^3$,

$$\omega(a \cdot v_p + w_p) = a \cdot \omega(v_p) + \omega(w_p) \text{ for all } a \in \mathbb{R} \text{ and } v_p, w_p \in T_p \mathbb{R}^3.$$

That is, the restriction $\omega_p: T_p \mathbb{R}^3 \rightarrow \mathbb{R}: v_p \mapsto \omega(v_p)$ is a linear functional.

Def. Let $\omega, \omega_1, \omega_2$ be 1-forms, and $f, g: \mathbb{R}^3 \rightarrow \mathbb{R}$ be functions, and $V, W: \mathbb{R}^3 \rightarrow \bigcup_{p \in \mathbb{R}^3} T_p \mathbb{R}^3$ be vector fields. Define the operations:

$$(\omega_1 + \omega_2)(v_p) \stackrel{\text{def}}{=} \omega_1(v_p) + \omega_2(v_p) \text{ and } (f\omega)(v_p) \stackrel{\text{def}}{=} f(p) \cdot \omega(v_p)$$

$$\omega(fV + gW) \stackrel{\text{def}}{=} f\omega(V) + g\omega(W) \text{ and } (f\omega_1 + g\omega_2)(V) = f\omega_1(V) + g\omega_2(V)$$

Def. Given a differentiable function $f: \mathbb{R}^3 \rightarrow \mathbb{R}$, the 1-form

$$df: \bigcup_{p \in \mathbb{R}^3} T_p \mathbb{R}^3 \rightarrow \mathbb{R}: v_p \mapsto v_p[f]$$

is called the differential of f .

Further, for the projections $\pi_i: \mathbb{R}^3 \rightarrow \mathbb{R}: (x_1, x_2, x_3) \mapsto x_i$ ($i=1, 2, 3$).

the set $\{(dx_1)_p, (dx_2)_p, (dx_3)_p\}$ is a dual basis for $T_p \mathbb{R}^3$ corresponding to $\{(e_1)_p, (e_2)_p, (e_3)_p\}$ for each $p \in \mathbb{R}^3$.

Lemma. Given 1-form ω on $\mathbb{R}^3$,

$$\omega = \sum_{i=1}^3 \omega \circ e_i \cdot dx_i.$$

In particular, $df = \sum_{i=1}^3 df \circ e_i \cdot dx_i = \sum_{i=1}^3 \frac{\partial f}{\partial x_i} \cdot dx_i$

Example.

Let $f(x, y, z) = (x^2 - 1)y + (y^2 + 2)z$. Then,

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz$$

$$= 2xy dx + (x^2 + 2yz - 1) dy + (y^2 + 2) dz.$$

So, $v_p[f] = df(v_p) = 2p_1 p_2 v_1 + (p_1^2 + 2p_2 p_3 - 1) v_2 + (p_2^2 + 2) v_3.$

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so $V_p[f] = df(V_p) = 2p_1 p_2 V_1 + (p_1^2 + 2p_2 p_3 - 1) V_2 + (p_2^2 + 2) V_3.$